

CHAMATH DE SILVA

chamathdesilva975@gmail.com | 226-962-0277

[GitHub/chamathdesilva1015](#) | [LinkedIn/Chamath De Silva](#) | [Technical Portfolio: desilva-chamath](#)

EDUCATION

Bachelor of Applied Science - Co-op (BASc)

Sept 2023 – Present

Level 3 - Honours Computer Science

McMaster University Hamilton, Ontario

Relevant Coursework: Operating Systems | Data Structures and Algorithms | Software Testing | Networks and Security

TECHNICAL SKILLS

Languages: C++ (Primary), Python (Tooling), SQL, Java, Haskell, JavaScript/TypeScript.

Systems & Infrastructure: Linux (Kernel/Shell), Multithreading (pthreads/std::thread), Memory, Docker, Git.

Performance & Tools: Profiling (perf, Valgrind), Benchmarking, GDB Debugging, CI/CD, REST APIs.

Technical Competencies: High-Concurrency Design, Performance Optimization, System Architecture, SDLC.

PROJECTS

Concurrent File Search Tool C++ | *Multithreading* | *Systems Programming* | *Performance Optimization* 2025 - 2026

- Engineered a high-performance search utility in C++17 utilizing a custom RAII-managed thread pool with condition variables and mutex-protected queues to achieve a 6.63x speedup on datasets of 1M+ lines.
- Optimized system I/O through non-blocking, recursive directory traversal and atomic result aggregation, reducing total search latency from 1114ms to 168ms.
- Implemented a task-stealing queue architecture to balance load across 10+ worker threads with zero race conditions.

Architect: Codebase Scaffolding Engine Python | *Jinja2* | *System Integration* | *CLI Development* 2026

- Developed a globally installable CLI tool that utilizes Jinja2 templating to generate performance-optimized, framework-agnostic architecture blueprints for C++, Python, and JavaScript.
- Integrated a "Founder's Vault" state management system using local JSON storage to track technical sketches and provide data-driven architectural rationale for developer decision-making.
- Engineered a Live-Serve automation layer using Python's concurrent execution modules to provide instant feedback loops from "Idea Sketch" to "Running Web App" in under 10 seconds.

Volatility Risk Engine Python | *Statistical Analysis* | *Data Engineering* | *Matplotlib* 2025 - 2026

- Implemented a high-efficiency hardware audit system using Z-Score analysis ($|Z| > 2$) to isolate price outliers and detect fraud in unstructured high-value datasets in real-time.
- Engineered a weighted risk matrix that dynamically classifies asset listings into 4-tier risk categories, generating automated audit trails and performance-trend visualizations.
- Developed an automated OOP pipeline featuring strict separation of concerns between data cleaning, statistical analysis, and visual reporting classes to ensure system extensibility.

WORK EXPERIENCE

Software Engineer Intern | *Callidora Trading Company Ltd - Guelph, ON*

May 2026 – Present

- Engineering scalable ETL/ELT data pipelines and production-quality backend services in Python and SQL to support high-concurrency machine learning and large-scale analytics workflows.
- Optimizing data ingestion and storage within Google Cloud Platform (GCP) to ensure high system performance, reliability, and automated validation for production datasets.
- Collaborating with cross-functional teams to translate business requirements into technical designs focused on horizontal scalability and long-term system maintainability.

CERTIFICATIONS

CS50x: Introduction to Computer Science | *Harvardx*

2026

CS50: Introduction to Cybersecurity | *Harvardx*

Feb 2026